Proof

By property 3, we have

$$\int_{-a}^{a} f(x)\,dx = \int_{-a}^{0} f(x)\,dx + \int_{0}^{a} f(x)\,dx.$$

In the integral $\int_{-a}^{0} f(x)\,dx$, let us make the substitution, $x = -u$. Then, $dx = -du$.

When $x = -a$, we get $u = a$, when $x = 0$, we get $u = 0$, So, we get

$$\int_{-a}^{0} f(x)\,dx = \int_{a}^{0} f(-u)(-du) = \int_{0}^{a} f(-u)\,du = \int_{0}^{a} f(-x)\,dx = \int_{0}^{a} f(x)\,dx. \quad \text{... (2)}$$

Substituting equation (2) in equation (1), we get

$$\int_{-a}^{a} f(x)\,dx = \int_{0}^{a} f(x)\,dx + \int_{0}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx.$$

Property 9

If $f(x)$ is an odd function, then $\int_{-a}^{a} f(x)\,dx = 0$.

(Recall that a function $f(x)$ is an odd function if and only if $f(-x) = -f(x)$.)

Proof

By property 3, we have

$$\int_{-a}^{a} f(x)\,dx = \int_{-a}^{0} f(x)\,dx + \int_{0}^{a} f(x)\,dx. \quad \text{... (1)}$$

Consider $\int_{-a}^{0} f(x)\,dx$. In this integral, let us make the substitution, $x = -u$. Then, $dx = -du$.

When $x = -a$, we get $u = a$; when $x = 0$, we get $u = 0$. So, we get

$$\int_{-a}^{0} f(x)\,dx = \int_{a}^{0} f(-u)(-du) = \int_{0}^{a} f(-u)\,du = \int_{0}^{a} f(-x)\,dx = -\int_{0}^{a} f(x)\,dx. \quad \text{... (2)}$$

Substituting equation (2) in equation (1), we get

$$\int_{-a}^{a} f(x)\,dx = \int_{0}^{a} f(x)\,dx - \int_{0}^{a} f(x)\,dx = 0$$

Property 10

If $f(2a - x) = f(x)$, then $\int_{0}^{2a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$.

Proof

By property 7, we have

$$\int_{0}^{2a} f(x)\,dx = \int_{0}^{a} \left[f(x) + f(2a - x)\right] dx. \quad \text{...(1)}$$

Setting the condition $f(2a - x) = f(x)$ in equation (1), we get

$$\int_{0}^{2a} f(x)\,dx = \int_{0}^{a} \left[f(x) + f(x)\right] dx = 2\int_{0}^{a} f(x)\,dx.$$

Property 11

If $f(2a - x) = -f(x)$, then $\int_{0}^{2a} f(x)\,dx = 0$.

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